Product Requirements Document (PRD)

MedFace – Facial Recognition-Based Patient History Management System

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Author: [Your Name / Team]

1. Purpose

MedFace is a secure, web-based medical application designed to enable healthcare professionals to quickly and accurately access a patient's medical history using facial recognition technology. This system aims to improve patient identification accuracy, reduce manual errors, and streamline medical record retrieval, ensuring timely and effective patient care.

2. Background

Current patient identification methods often rely on manual input or ID cards, which can lead to errors, delays, or privacy concerns. Leveraging facial recognition technology provides a seamless, contactless, and efficient way to identify patients and access their medical history instantly.

3. Scope

- Web application accessible by doctors, nurses, receptionists, and optionally patients.
- Facial recognition-based patient identification.
- Secure storage and management of medical records.
- Role-based access control.
- Upload and management of medical reports.
- Search and filter functionality for medical records.

4. Stakeholders

Role	Description
Doctors	Primary users accessing patient data
Nurses	Support medical staff with limited access
Receptionists	Manage patient enrollment and appointments
Patients	Optional access to their own medical data
Admins	Manage users, roles, and system settings

5. Functional Requirements

5.1 User Authentication & Authorization

- Users must log in via secure authentication (JWT/OAuth).
- Role-based access control (Doctor, Nurse, Receptionist, Admin, Patient).
- Session management and logout functionality.

5.2 Patient Face Enrollment

- Capture patient face image via webcam or mobile camera.
- Extract face embeddings using FaceNet/OpenCV.
- Store embeddings securely linked with patient ID.
- Allow manual patient data entry during enrollment.

5.3 Patient Identification

- Doctor initiates face scan.
- System captures live image, generates embedding.
- Embedding compared against stored vectors in vector DB.
- On match, patient medical history is retrieved and displayed.

5.4 Medical History Management

- View patient demographics, allergies, chronic conditions.
- Display prescriptions, diagnostics, visit logs.
- Add/edit prescriptions and diagnoses (doctor role).
- Upload and tag reports (PDFs, images, lab results).

5.5 Search & Filter

- Search patients by name, date, symptoms, doctor.
- Filter medical records by date range, type, or tags.

5.6 Admin Panel

- Add/remove/manage users (doctors, nurses, receptionists).
- Assign roles and permissions.
- View access logs and audit trails.

5.7 Patient Panel (Optional)

- Patients can view their own medical history.
- Download reports and prescriptions.

6. Non-Functional Requirements

6.1 Security

- Encrypt all medical data at rest and in transit (TLS, AES-256).
- Do not store raw facial images; store only embeddings.
- Maintain audit logs of all access and changes.
- Comply with relevant data privacy laws (HIPAA, GDPR).

6.2 Performance

- Face recognition response time < 2 seconds.
- System uptime ≥ 99.5%.

6.3 Scalability

- Support growing number of users and patient records.
- Modular architecture to add biometric options or mobile apps.

6.4 Usability

- Intuitive UI for doctors and staff.
- Responsive design for desktops and tablets.

7. Technical Architecture

- Frontend: React.js / Next.js
- Backend: Node.js (Express) or Python (FastAPI)
- Database: PostgreSQL or MongoDB
- Face Recognition: FaceNet / OpenCV + Dlib
- Vector Search: Faiss or Milvus for embedding matching
- Storage: AWS S3 or Firebase Storage for reports
- Hosting: Vercel (frontend), Render/Railway/VPS (backend)

8. Milestones & Timeline

Week	Deliverable
1	UI/UX Design, Database Schema
2-3	Frontend Development (Login, Dashboard, Face Scan)
4-5	Backend API & Database Setup
6-7	Face Recognition Integration
8	Medical Record CRUD & Reports Module
9	Security Hardening, Role Management, Testing
10	Deployment, QA, Bug Fixes

9. Success Metrics

- Accurate patient identification rate ≥ 95%.
- Average face recognition time ≤ 2 seconds.
- User satisfaction score ≥ 85% (via surveys).
- Zero data breaches or privacy incidents.
- System uptime ≥ 99.5%.

10. Risks & Mitigation

Risk	Mitigation
Privacy concerns over facial data	Store only embeddings, encrypt data
False positives in recognition	Use threshold tuning, manual override option
Regulatory compliance	Consult legal experts, implement best practices
System downtime	Use reliable hosting, implement monitoring

11. Future Enhancements

- Add fingerprint or other biometric authentication.
- Develop patient mobile app for remote access.
- Integrate telemedicine features.
- Connect prescriptions directly to pharmacies.

12. Budget Considerations

- Utilize free tiers of cloud services (Firebase, MongoDB Atlas).
- Open-source libraries for face recognition.
- Cost-effective hosting solutions (Vercel, Render).
- Budget allocation for security audits and compliance.

Appendix

- Glossary of terms
- References to HIPAA, GDPR guidelines
- Links to FaceNet, OpenCV, Faiss documentation

If you need, I can help you draft detailed user stories, wireframes, or technical specifications next!